MSE 5024 Advanced Thermodynamics and Kinetics

HOMEWORK 8

Name: *Student ID:*

Please hand in this homework before class on April 19th, 2022.

1. The vapor pressure of solid NaF varies with temperature as

$$\ln p(atm) = \frac{-34,450}{T} - 2.01\ln T + 33.74$$

and the vapor pressure of liquid NaF varies with temperature as

$$\ln p(atm) = \frac{-31,090}{T} - 2.52\ln T + 34.66$$

Calculate

a) The normal boiling temperature of NaF;

b) The temperature and pressure at the triple point;

c) The molar enthalpy of evaporation of NaF at its normal boiling temperature;

d) The molar enthalpy of melting of NaF at the triple point;

e) The difference between the constant-pressure molar heat capacities of liquid and solid NaF.

1. The normal boiling temperature, T_b, is defined as that temperature at which the saturated vapor pressure of the liquid is 1 atm. Thus, from the equation for the vapor pressure of the liquid, T_b is

$$\ln(1) = 0 = -\frac{31,090}{T_b} - 2.52\ln T_b + 34.66$$

which has the solution

$$T_b = 2006 \text{ K}$$

2. The saturated vapor pressures for the solid and liquid phases intersect at the triple point. Thus, at the temperature, T_{tp}, of the triple point

$$-\frac{34,450}{T_{tp}} - 2.01\ln T_{tp} + 33.74 = -\frac{31,090}{T_{tp}} - 2.52\ln T_{tp} + 34.66$$

which has the solution

$$T_p = 1239 \text{ K}$$

The triple-point pressure is then calculated from the equation for the vapor pressure of the solid as

$$p = \exp\left(-\frac{34,450}{1239} - 2.01\ln 1239 + 33.74\right) = 2.29\times10^{-4} \text{ atm}$$

or from the equation for the vapor pressure for the liquid as

$$p = \exp\left(-\frac{31,090}{1239} - 2.52\ln 1239 + 34.66\right) = 2.29\times10^{-4} \text{ atm}$$

3. For vapor in equilibrium with the liquid:

$$\ln p\ (\text{atm}) = -\frac{31,090}{T} - 2.52\ln T + 34.66$$
$$\frac{d\ln p}{dT} = \frac{\Delta H}{RT^2} = \frac{31,090}{T^2} - \frac{2.52}{T}$$

Thus,

$$\Delta H_{(l\to v)} = (31,090\times8.3144) - (2.52\times8.3144)T = 258,500 - 20.95\text{T}$$

and, at the normal boiling temperature of 2006 K,

$$\Delta H_{(l\to v)} = 258,500 - 20.95\times2006 = 216,500 \text{ J}$$

4. For vapor in equilibrium with the solid:

$$\ln p\ (\text{atm}) = -\frac{34,450}{T} - 2.01\ln T + 33.74$$

Thus,

$$\begin{aligned}\Delta H_{(s\to v)} &= (34,450\times 8.3144) - (2.01\times 8.3144)T \\ &= 286,400 - 16.71T \text{ J}\end{aligned}$$

Near the triple point,

$$\Delta H_{(s\to l)} + \Delta H_{(l\to v)} = \Delta H_{(s\to v)}$$

and thus,

$$\begin{aligned}\Delta H_{(s\to l)} &= 286,400 - 16.71T - 258,500 \times 20.95T \\ &= 27,900 + 4.24T\end{aligned}$$

At the triple point,

$$\Delta H_{(s\to l)} = 27,900 + (4.24\times 1239) = 33,150 \text{ J}$$

5 $\Delta H_{(s\to l)} = 27,900 + 4.24\text{T}$

$$\begin{aligned}\frac{d\Delta H}{dT} &= \Delta c_p = 4.24 \text{ J/K} \\ &= c_P^L - c_P^S > 0\end{aligned}$$

2. Carbon has the following three allotropes: graphite, diamond, and a metallic form called solid III. Calculate the pressure which, when applied to one mole graphite at 1800 K, causes the transformation of graphite to diamond, given

$H_{graphite} - H_{diamond} = -1896J$

$S_{graphite} = 5.74J / K$

$S_{diamond} = 2.36J / K$

Density of graphite: 2.62 g/cm^3

Density of diamond:3.52 g/cm^3

For the transformation graphite → diamond at 1800 K:

$$\Delta G = \Delta H - T\Delta S = 1896 - 1800\times(2.36 - 5.74) = 7980J$$

For the transformation of graphite to diamond at any temperature T:

$$\Delta V = V_{diamond} - V_{graphite} = \frac{12}{3.52} - \frac{12}{2.62} = -1.17cm^3 / mole$$

Equilibrium between graphite and diamond at 298 K requires that $\Delta G_{graphite \to diamond}$ be zero. Since

$$\Delta G(P, T = 1800) = \Delta G(P = 1, T = 1800) + \int_1^P \Delta V dP$$

If the difference between the isothermal compressibilities of the two phases is negligibly small (i.e., if the influence of pressure on ΔV can be ignored), then

$$-7980 = -1.17 \times 0.1013(P - 1)$$

$$P = \frac{7980}{1.17 \times 0.1013} + 1 = 67314atm$$

3. You are responsible for the purchase of oxygen gas which, before use, will be stored at a pressure of 200 atm at 300 K in a cylindrical vessel of diameter 0.2 m and height 2 m. Would you prefer that the gas behaved ideally or as a van der Waals fluid? The van der Waals constants for oxygen are a = 1.36 liters^2·atm·mole^{-2} and b = 0.0318 liters/mole.

$$v = \frac{\pi d^2}{4} h = \frac{1}{4} \times 3.14 \times 0.2^2 \times 2 = 0.0628m^3$$

the gas behaved ideally:

$$n = \frac{Pv}{RT} = \frac{200 \times 10125 \times 0.0628}{8.314 \times 300} = 510mole$$

the gas behaved as a van der Waals fluid:

$$\left(P + \frac{n^2 a}{v^2}\right)(v - nb) = nRT$$

$$n = 565mole$$

The gas behaved as a van der Waals fluid is better.

4. Derive the reduced equation of state for the Pedlich-Kwong's equation.

$$P=\frac{RT}{(v-b)}-\frac{a}{\sqrt{T}v(v+b)}$$

$$\begin{cases}\left(\frac{\partial P}{\partial v}\right)_{T_{cr}}=-\frac{RT_{cr}}{(v_{cr}-b)^2}+\frac{a(2v_{cr}+b)}{\sqrt{T_{cr}}v_{cr}^{\ 2}(v_{cr}+b)^2}=0\\ \left(\frac{\partial^2 P}{\partial v^2}\right)_{T_{cr}}=\frac{2RT_{cr}}{(v_{cr}-b)^3}+\frac{a}{\sqrt{T_{cr}}}\frac{2v_{cr}^{\ 2}(v_{cr}+b)^2-2(2v_{cr}+b)^2v_{cr}(v_{cr}+b)}{v_{cr}^{\ 4}(v_{cr}+b)^4}=0\end{cases}$$

$$2v_{cr}^{\ 2}(v_{cr}+b)^2-2(2v_{cr}+b)^2v_{cr}(v_{cr}+b)=-6v_{cr}^{\ 4}-8v_{cr}^{\ 3}b-4v_{cr}^{\ 2}b^2-2v_{cr}b^3$$

$$\frac{a(2v_{cr}+b)}{\sqrt{T_{cr}}v_{cr}^{\ 2}(v_{cr}+b)^2}=\frac{RT_{cr}}{(v_{cr}-b)^2}\Rightarrow T_{cr}=\left[\frac{a(2v_{cr}+b)(v_{cr}-b)^2}{Rv_{cr}^{\ 2}(v_{cr}+b)^2}\right]^{\frac{2}{3}}$$

$$\frac{2RT_{cr}}{(v_{cr}-b)^3}=\frac{a}{\sqrt{T_{cr}}}\frac{6v_{cr}^{\ 4}+8v_{cr}^{\ 3}b+4v_{cr}^{\ 2}b^2+2v_{cr}b^3}{v_{cr}^{\ 4}(v_{cr}+b)^4}\Rightarrow T_{cr}=\left[\frac{a(v_{cr}-b)^3}{RT_{cr}}\frac{6v_{cr}^{\ 4}+8v_{cr}^{\ 3}b+4v_{cr}^{\ 2}b^2+2v_{cr}b^3}{v_{cr}^{\ 4}(v_{cr}+b)^4}\right]^{\frac{2}{3}}$$

$$\frac{a(2v_{cr}+b)(v_{cr}-b)^2}{Rv_{cr}^{\ 2}(v_{cr}+b)^2}=\frac{a(v_{cr}-b)^3}{R}\frac{6v_{cr}^{\ 4}+8v_{cr}^{\ 3}b+4v_{cr}^{\ 2}b^2+2v_{cr}b^3}{v_{cr}^{\ 4}(v_{cr}+b)^4}$$

$$(2v_{cr}+b)v_{cr}^{\ 2}(v_{cr}+b)^2=(v_{cr}-b)(6v_{cr}^{\ 4}+8v_{cr}^{\ 3}b+4v_{cr}^{\ 2}b^2+2v_{cr}b^3)$$

$$2v_{cr}^{\ 5}+5v_{cr}^{\ 4}b+4v_{cr}^{\ 3}b^2+v_{cr}^{\ 2}b^3=6v_{cr}^{\ 5}+2v_{cr}^{\ 4}b-4v_{cr}^{\ 3}b^2-2v_{cr}^{\ 2}b^3-2v_{cr}b^4$$

$$4v_{cr}^{\ 5}-3v_{cr}^{\ 4}b-8v_{cr}^{\ 3}b^2-3v_{cr}^{\ 2}b^3-2v_{cr}b^4=0$$

阶次太高，无法求解

5. Look for the definition or derivation of Eulerian stress as follows and give the corresponding literature.

$$f=\frac{1}{2}\left[\left(\frac{V_0}{V}\right)^{2/3}-1\right]$$